Task 1: Library Book Lending

BCS | OOP | 40 minutes | Instructor: Dr Muhammad Shahid Bhatti

Read the scenario, identify classes, sketch UML, then code.

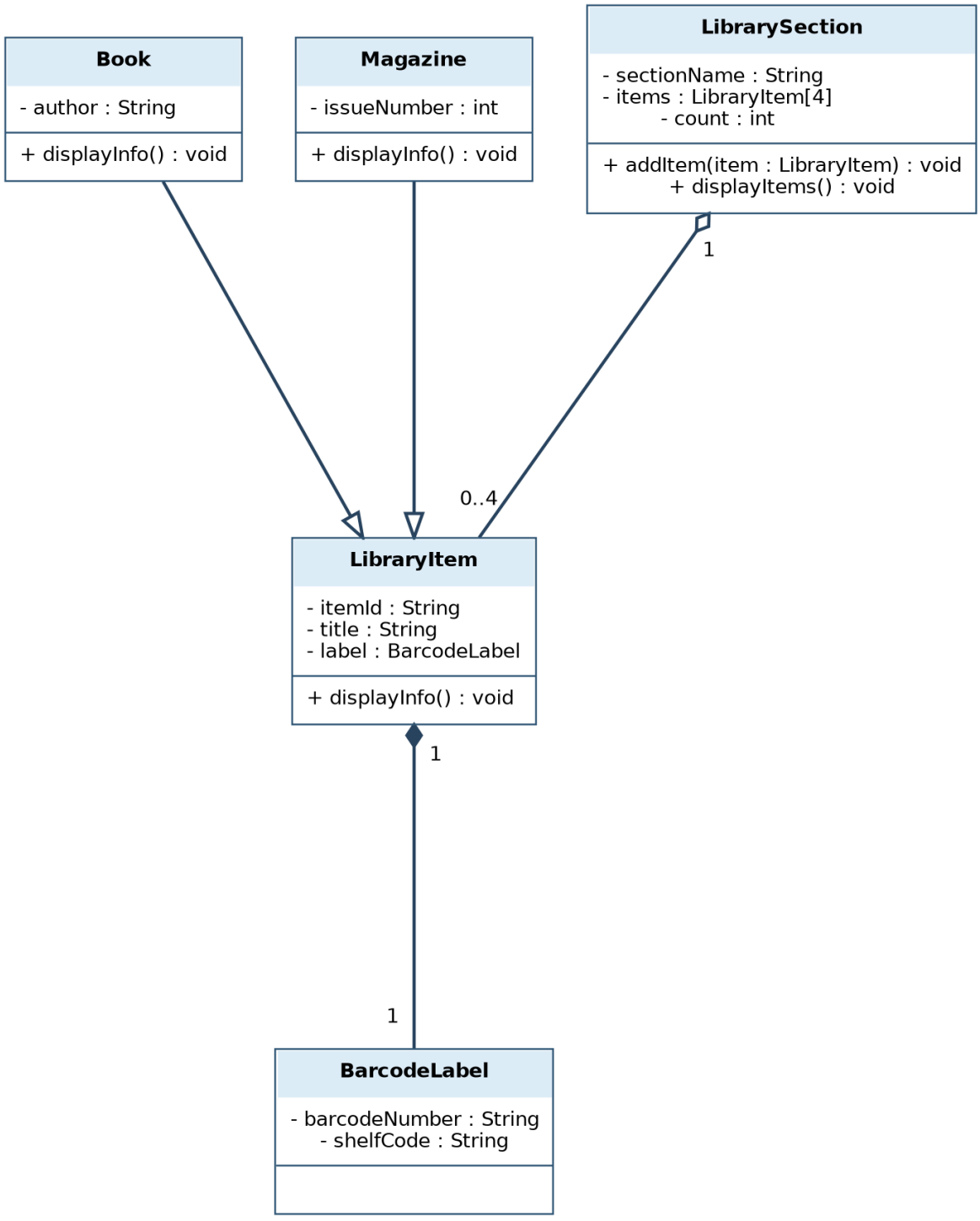
Scenario

A small college library keeps books and magazines on its shelves. Each library item has an item ID and a title. A Book also stores its author, while a Magazine stores its issue number. Since both are library items, they should reuse common properties and behavior from a parent class.

Each library item has its own BarcodeLabel, containing a barcode number and a shelf code. The barcode label is prepared for that particular item when the item object is created. The program must create the BarcodeLabel inside the LibraryItem constructor; do not pass an already-created label object.

The library also has a LibrarySection, for example “Computing Section.” A section can hold up to four existing library items. Books and magazines are created first and then added to the section. The items should continue to exist even if they are not added to any section.

UML Class Diagram



Read the UML notation: hollow triangle = inheritance; black diamond = composition; white diamond = aggregation; plain line = association. Multiplicities show how many objects participate. The editable PlantUML source is provided separately.

Classes and Attributes

Class	Required attributes
LibraryItem	itemId, title, BarcodeLabel label

Book extends LibraryItem	author
Magazine extends LibraryItem	issueNumber
BarcodeLabel	barcodeNumber, shelfCode
LibrarySection	sectionName, LibraryItem[] items, count

Required Methods

- LibraryItem.displayInfo(): show common item fields and barcode details.
- Book.displayInfo() and Magazine.displayInfo(): override and add their specific fields; call super.displayInfo().
- LibrarySection.addItem(LibraryItem item): add if the fixed array is not full.
- LibrarySection.displayItems(): display all stored items using a loop.

What to Create in main()

- Create two Book objects and one Magazine object.
- Create one LibrarySection with capacity four; add all three existing items.
- Print the section name and information of all items.
- Create two more items; fill the section with four items, then try adding the fifth and show "Section full".

Relationships to Demonstrate

Inheritance: Book and Magazine extend LibraryItem. Composition: LibraryItem creates and owns BarcodeLabel. Aggregation: LibrarySection holds references to previously created LibraryItem objects.

Completion Checklist

- ☐ All classes compile and run.
- ☐ Constructors use super(...) in subclasses.
- ☐ Inherited display methods are overridden where specified.
- ☐ Composition objects are constructed within their owner.
- ☐ Aggregation objects are created first and added/passed afterward.
- ☐ The program runs and prints understandable output.

Submission

Save all .java source files for this task in its own folder. At the end of the lab, submit ONE ZIP file containing completed folders for Tasks 1–4. Do not submit four separate ZIP files.